Total Spending and Level of Economic Activity

Recovery or Expansion

Increased Spending

↓

Causes

↓

Larger Output

↓

Causes

↓

Increased Employment

↓

Causes

↓

More Income

Recession or Contraction

Decreased Spending

↓

Causes

↓

Smaller Output

↓

Causes

↓

Decreased Employment

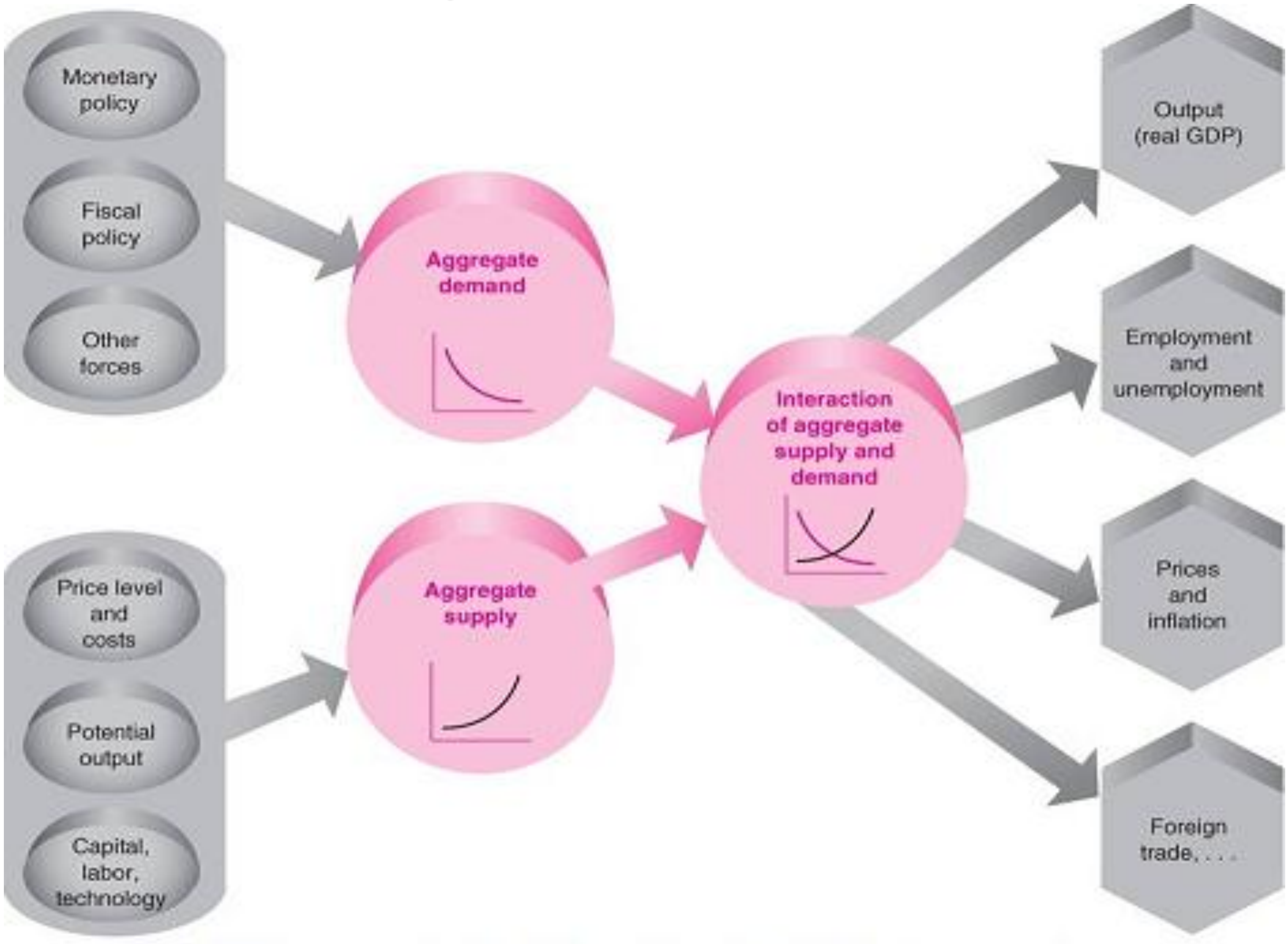
↓

Causes

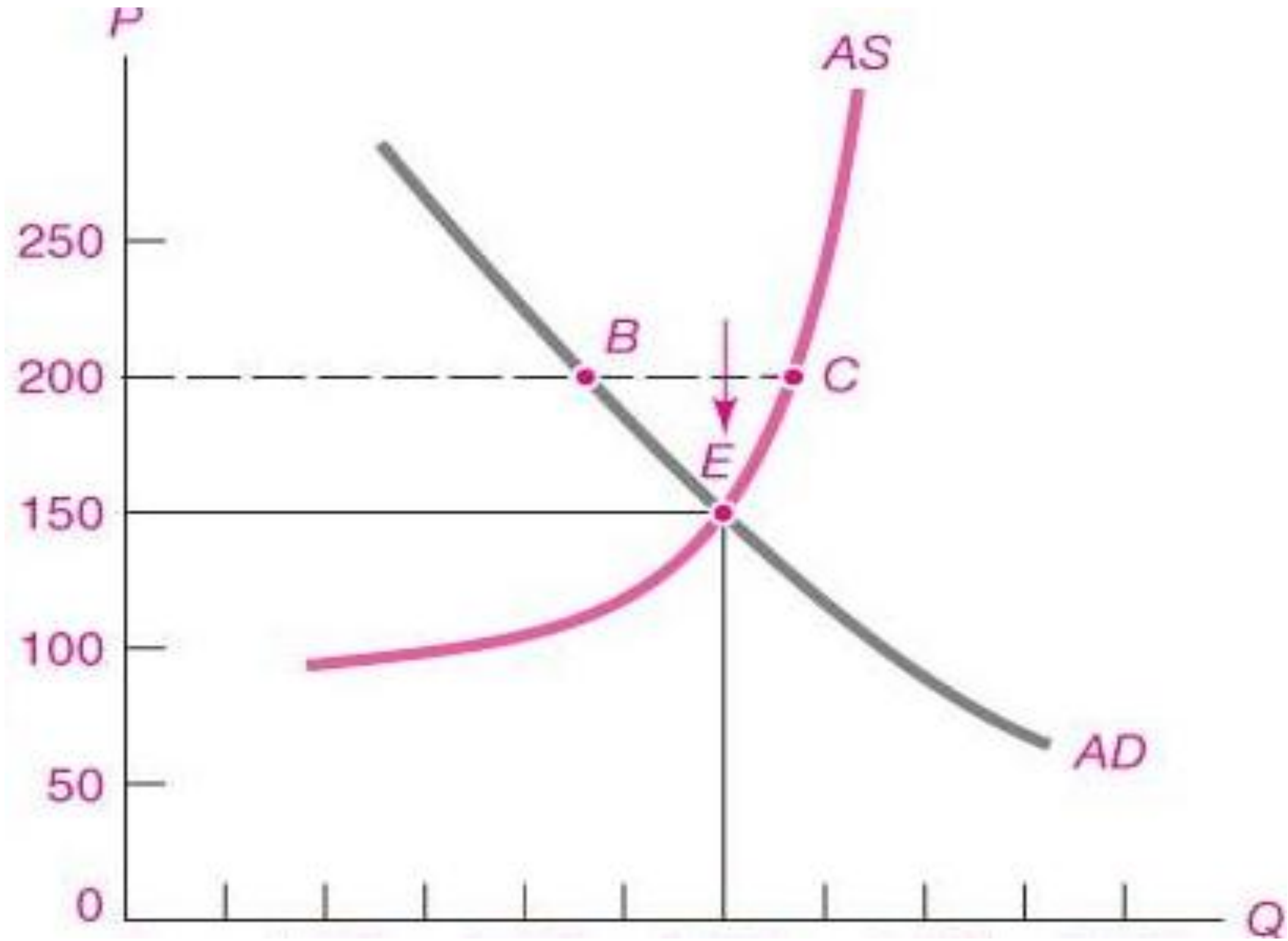
↓

Less Income

Macroeconomic System and Determinants of AD and AS

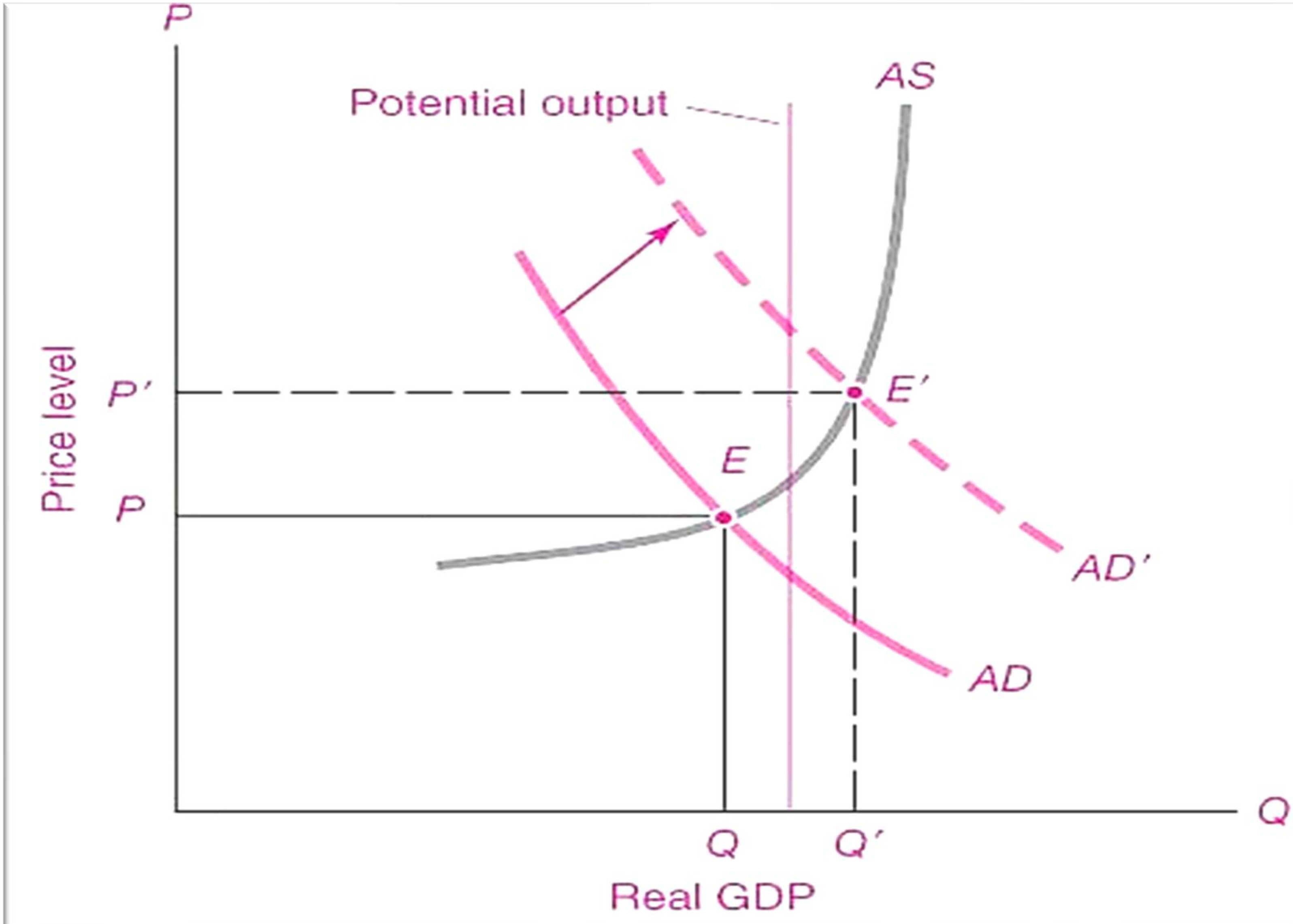


MACROECONOMIC EQUILIBRIUM



Equilibrium determination of National Income and general price level

Output can rise above the trend (or potential output) because people work overtime and machinery is used for several shifts.



Measuring Joblessness: The Unemployment Rate

Labor Force = Number of Employed + Number of Unemployed

$$\text{Unemployment Rate} = \frac{\text{Number of Unemployed}}{\text{Labor Force}} \times 100$$

$$\text{Labor-Force Participation Rate} = \frac{\text{Labor Force}}{\text{Adult Population}} \times 100$$

INVESTMENT

The second largest component of aggregate demand in an economy after consumption, is investment.

Two important roles:

Large and highly volatile component of aggregate demand affecting the business cycle.

Investment leads to capital accumulation and promotes economic growth in the long run.

The most volatile component of GDP over the business cycle.

INVESTMENT

- Business Fixed Investment
- Residential Investment
- Inventory Investment

Business fixed investment comprises mainly of business spending on machinery, equipment, and structures such as factories.

Residential investment consists largely of investment in housing and real estate.

Inventory investment considers only additions to the stock of inventories of firms: both final goods' inventories and inventories of raw materials.

Business Fixed Investment

Gross investment expenditures of a firm are conceptually broken up into two parts: net investment expenditures on *new* capital goods and replacement investment expenditures on the *depreciated* stock of capital. The latter includes both maintenance expenditures on usable capital inherited from the past, and expenditures to replace the obsolete capital goods.

MARGINAL EFFICIENCY OF CAPITAL

Investment, in the theory of income and employment, means an addition to the nations physical stock of capital like building of new factories, new machines as well as any addition to the stock of finished goods or the goods in the pipeline of production.

Investment has to create income and employment.

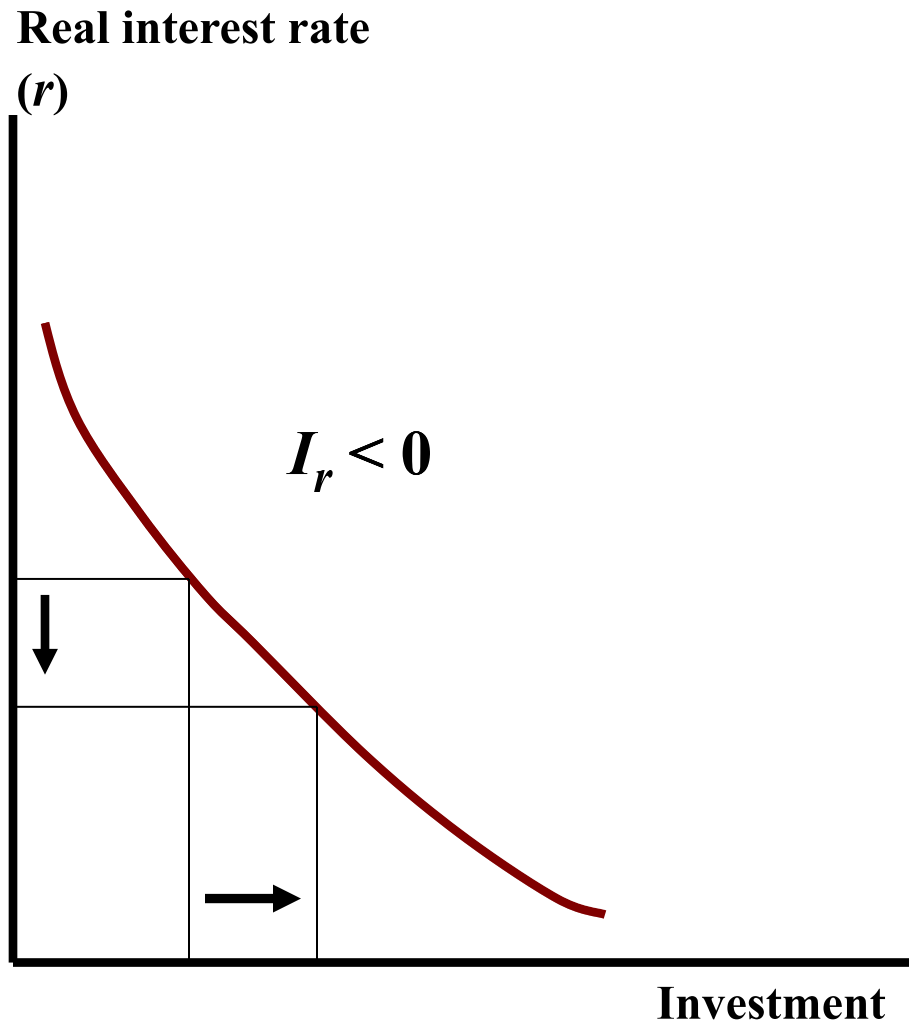
Marginal efficiency of investment is the highest expected rate of profit which is likely to be had by a marginal increase in the rate of investment.

MEC must never fall below the current rate of interest, if investment is to be worthwhile.

The Profitability of Investment Depends on the Interest rate

Project	Investment (in Million $)	Annual Rev. per $1000	Cost per $1000 Borrowed		Net Profit	
			@10%	@5%	@10%	@5%
I	1	1500	100	50	1400	1450
II	4	220	100	50	120	170
III	10	160	100	50	60	110
IV	10	130	100	50	30	80
V	5	110	100	50	10	60
VI	15	90	100	50	–10	40
VII	10	60	100	50	–40	10
VIII	20	40	100	50	–60	–10

If net profit is positive, profit-maximizing firms will undertake the investment; if negative, the investment project will be rejected.



Downward Sloping Investment Function

Shifts in the Aggregate Demand Curve

- The downward slope of the aggregate demand curve shows that a fall in the price level raises the overall quantity of goods and services demanded.

When one of these other factors changes, the aggregate demand (AD) curve shifts.

Shifts arising from —

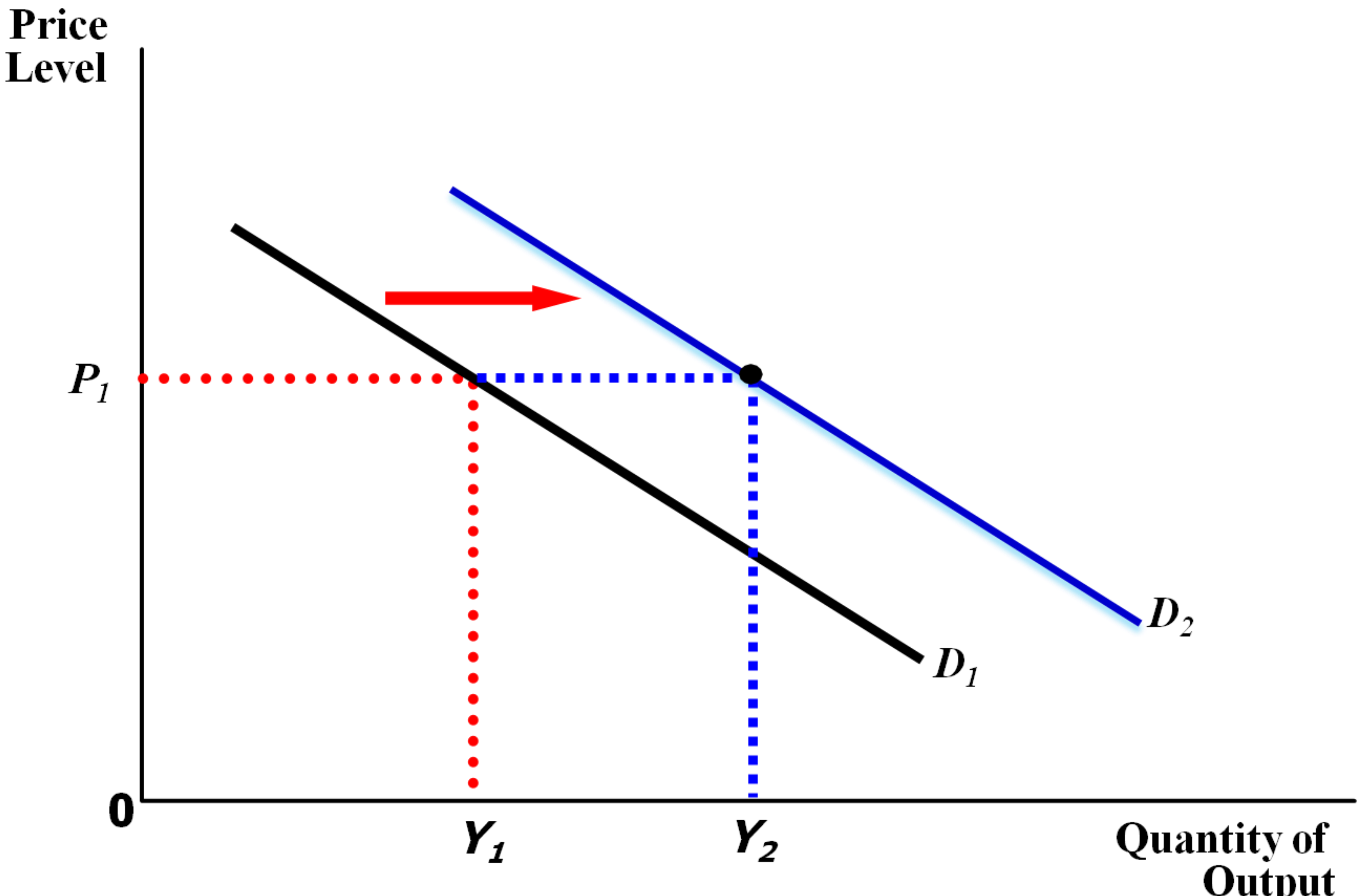
Consumption

Investment

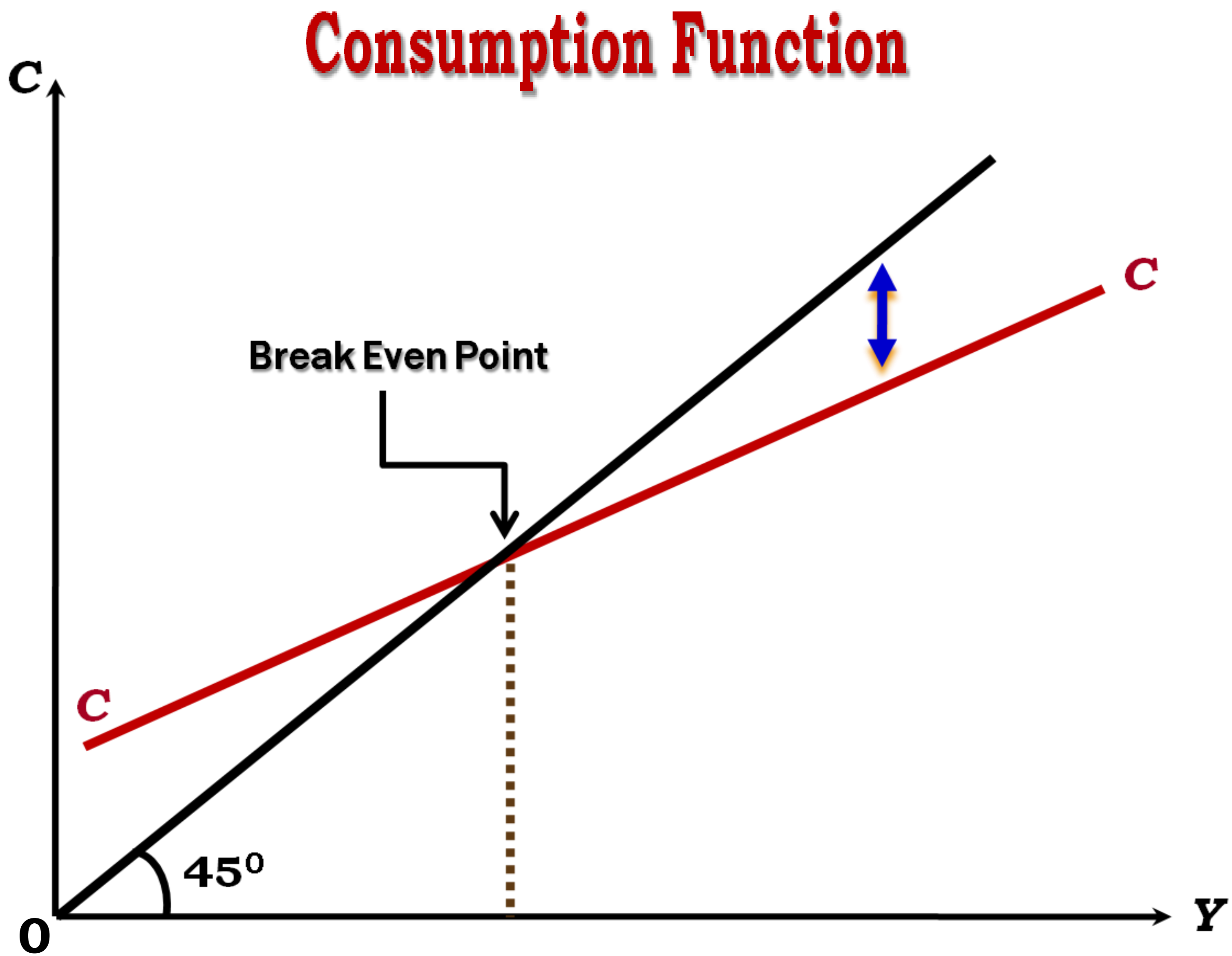
Government Purchases

Net Exports

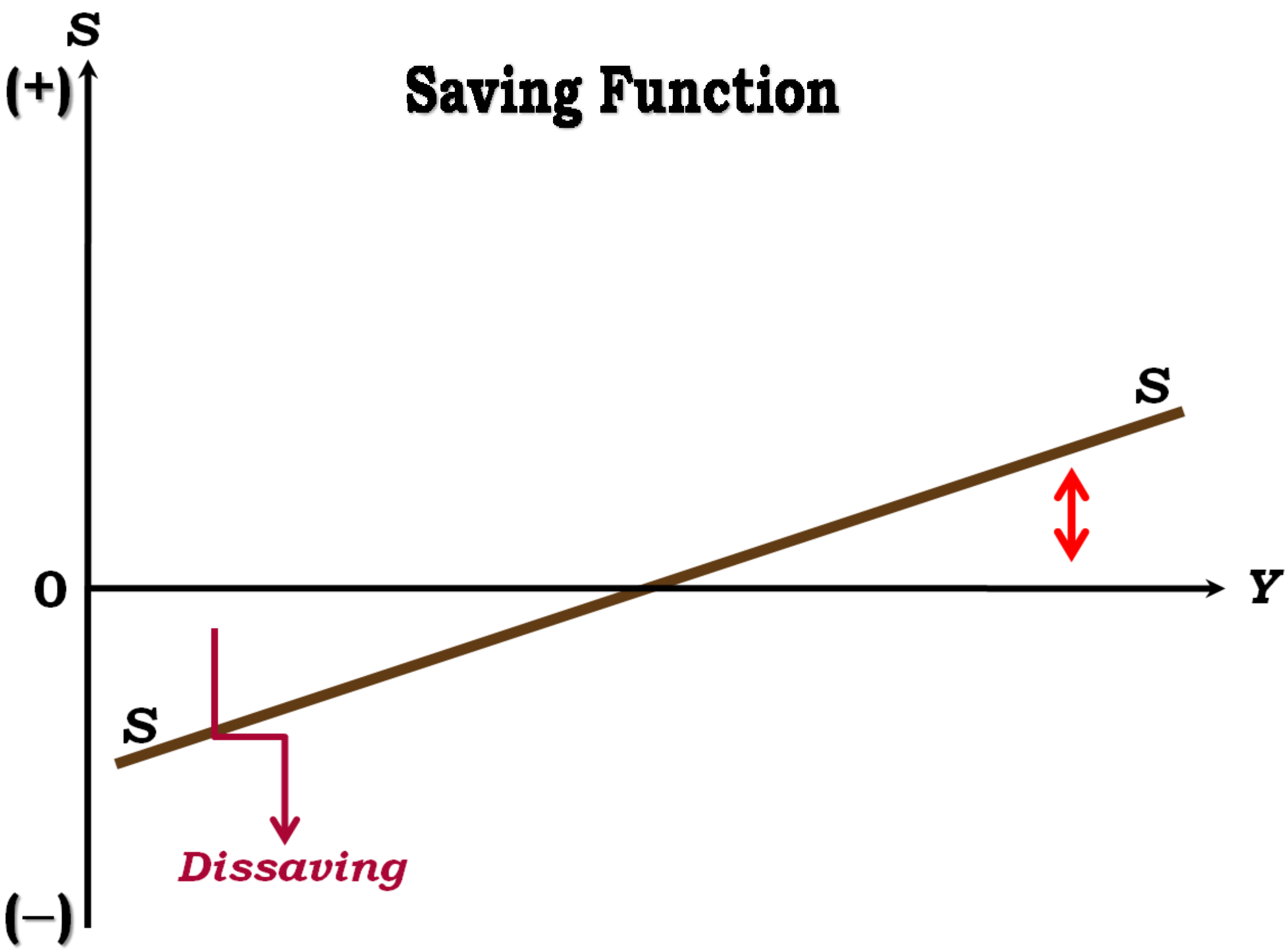
Shifts in the Aggregate Demand Curve



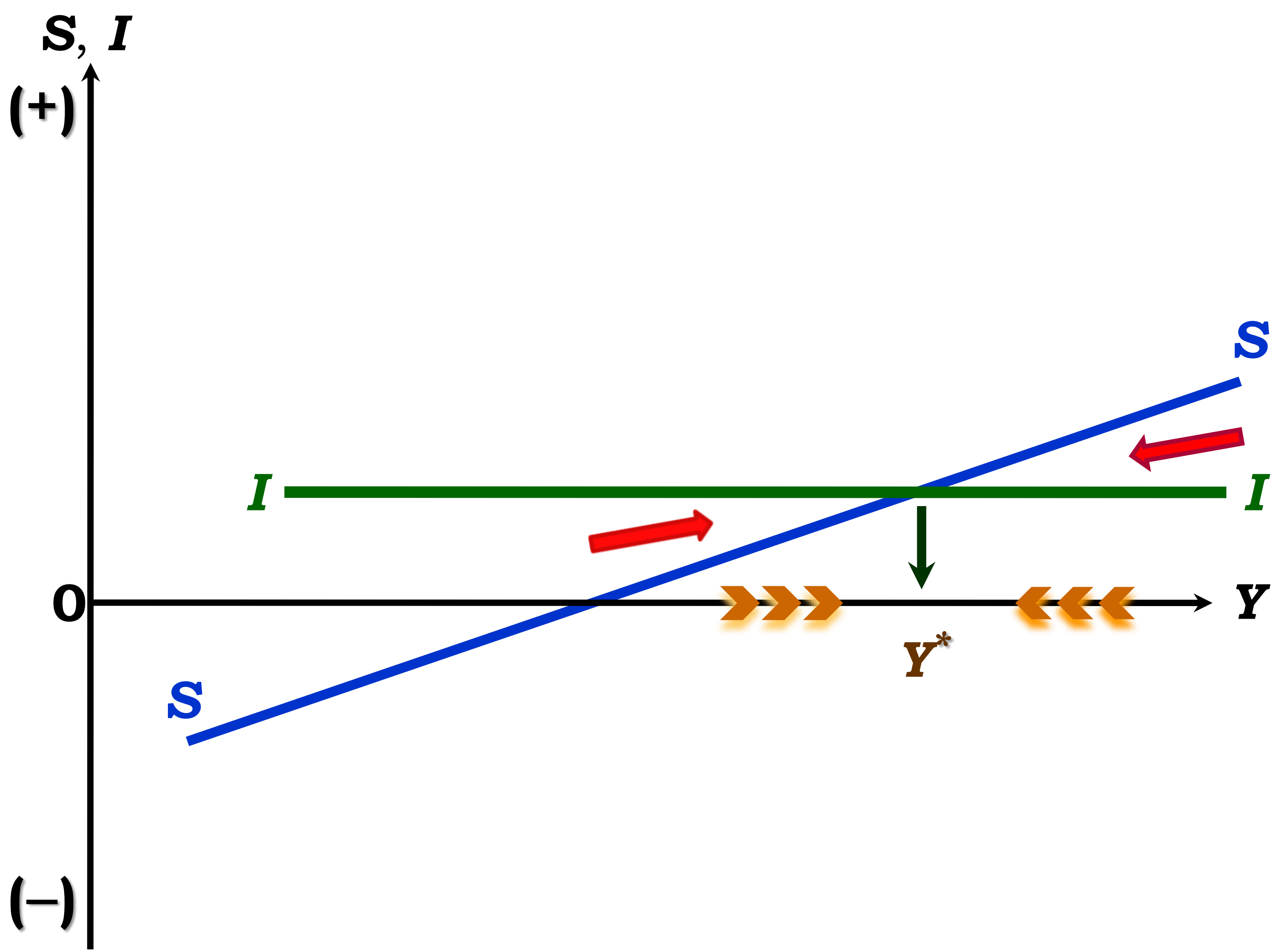
National Output Determines the Level of Consumption in the Economy



National Output Determines the Level of Savings in the Economy

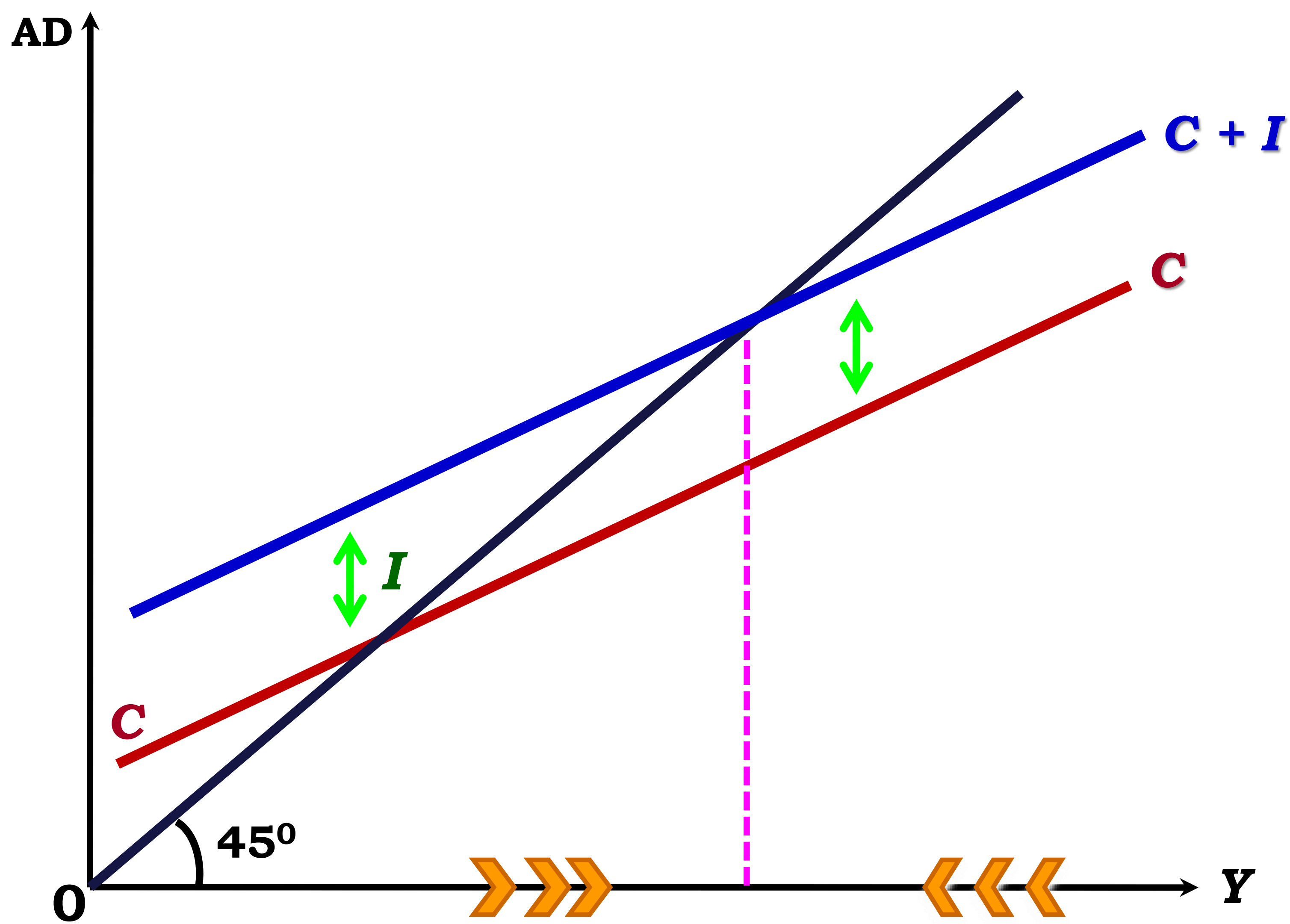


Saving & Investment Functions



The Equilibrium Output Is Determined by Intersection of S & I

Total Expenditure / Aggregate Demand



THE INVESTMENT MULTIPLIER

$$Y = C(Y) + \tilde{I}$$

Suppose, $\tilde{I}$ increases by 1 unit ⟶ Y increases by 1 unit

Consumption is a rising function of income, so C increases by C' unit

With increase in consumption, income will also rise by C' unit

Consumption further rises by $(C')^2$ unit ⟶ Income will rise by $(C')^2$ unit

With increase in income, consumption will again rise by $(C')^3$ unit

With increase in consumption, income will also rise by $(C')^3$ unit

... and this process continues.

$$\Delta Y = \{1 + (C') + (C')^2 + (C')^3 + \ \dots\ \}\Delta\tilde{I}$$

$$\frac{\Delta Y}{\Delta\tilde{I}} = \frac{1}{(1 - C')} \rangle 1$$

THE G - MULTIPLIER

Y = C (Y) + I + G

Suppose, G increases by 1 unit → Y increases by 1 unit.

Consumption is a rising function of income, so C increases by $\acute{C}$ unit.

With increase in consumption, income will also rise by $\acute{C}$ unit.

Consumption further rises by $(\acute{C})^2$ unit → Income will rise by $(\acute{C})^2$ unit

With increase in income, consumption will again rise by $(\acute{C})^3$ unit.

With increase in consumption, income will also rise by $(\acute{C})^3$ unit.

… and this process continues.

$$\Delta Y = \{1 + (C') + (C')^2 + (C')^3 + \ \dots \}\Delta G$$

$$\frac{\Delta Y}{\Delta G} = \frac{1}{(1 - C')} \rangle 1$$

The Balanced Budget Multiplier

Additional govt. expenditure is financed by additional taxation

$$Y = C\,(Y - T) + \tilde{I} + G$$

Assume that investment is autonomous and fixed

$$dY = \acute{C}\,dY - \acute{C}\,dT + d\tilde{I} + dG \qquad d\tilde{I} = 0$$

$$dT = dG$$

$$(1 - \acute{C})\,dY = (1 - \acute{C})\,dG$$

$$\left.\frac{dY}{dG}\right|_{dG=dT} = \frac{(1-C')}{(1-C')} = 1$$

THE TAX MULTIPLIER

Disposable Income = Income *minus* Taxes

Increase in Tax *reduces* the Disposable Income.

But Consumption is a rising function of Disposable Income.

With rise in tax, DI will be falling and so would be Consumption demand.

Consumption demand comprises the single largest portion of aggregate demand in the economy.

Hence, AD would fall.

Summary: The Autonomous Multipliers

Larger the MPC, greater the multiplier effect; but larger the MPS, smaller the multiplier effect.

Closed economy multiplier is greater than the open economy multiplier.

Investment and G-multipliers are *positive* but the tax multiplier is *negative*.